Yunui Kang

B.S. Candidate in Energy Engineering, KENTECH, Korea

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Research Interests

Security of energy cyber-physical systems: protocol flaws and attack detection in smart grids and EV charging.

Education

Korea Institute of Energy Technology (KENTECH) — Mar 2023 – Feb 2027 (Expected)
B.S. in Energy Engineering (Concentration: Electrical and Electronics Engineering) — *GPA 4.16/4.30*

- **Relevant Coursework:** Control Theory and Applications, Semiconductor Devices, Electronic Circuits, Circuit Theory, Network Science, Computer Architecture and Systems, Optimization Theory, Next-Generation Grid, HarvardX CS50's Introduction to Cybersecurity (transfer credit)

University of California, Los Angeles (UCLA) — Jun – Aug 2024
Summer Study Abroad Program — *Los Angeles, CA*

Hankuk Academy of Foreign Studies — Feb 2022
Yongin, Korea

Research Experience

The University of Texas at Dallas, System and Network Security Lab — Sep 2026 – Present
Visiting Researcher (KENTECH Global Research Program), Advisor: Prof. Imtiaz Karim — *Richardson, TX*

- Fuzzing two commercial EV chargers over the Open Charge Point Protocol (OCPP 1.6) to uncover deviations from the specification.
- Leading an LLM-agent pipeline that parses the OCPP specification, extracts security requirements, generates test cases, and judges charger responses.
- Adapting specification-based protocol analysis techniques from cellular network security to EV charging infrastructure.

KENTECH Computer System Security Lab — Mar 2025 – Present
Undergraduate Researcher, Advisor: Prof. Hyunwoo Lee — *Naju, Korea*

- Contributed to government-funded projects on AI-based multi-layer security and post-quantum cryptography (PQC) migration.
- **HD-NIDSBench (first author):** built a framework that benchmarks hyperdimensional computing (HDC) intrusion detection for resource-constrained IoT edge devices.
- Found that retraining improved F1-score by 17% on average over single-pass training, and that HDC used $\sim$80$\times$ less memory than a feed-forward network of similar F1-score (2.44 vs. 193 KB).
- **PQNetSim (second author):** testbed emulating lossy IoT networks with Linux tc and a Raspberry Pi client; ML-KEM added 59.9–373.7 ms to DTLS 1.3 handshakes and was estimated to lower handshake success rates by over 10%.

KAIST Graduate School of AI, Intelligent Network Architecture (INA) Lab — Jul – Aug 2026
Research Intern, Advisor: Prof. Dongsu Han — *Seoul, Korea*

- Studied GPU underutilization in agentic LLM serving, where GPUs sit idle during tool calls while KV caches stay resident in GPU memory.
- Implemented CPU-offloading schedulers aimed at improving GPU throughput under memory pressure.

Electronics and Telecommunications Research Institute (ETRI) — Jan – Feb 2026
Research Intern, AI Convergence Security Research Section — *Daejeon, Korea*

- Evaluated pseudo-labeling strategies with K-means, DBSCAN, and HDBSCAN clustering for a person re-identification benchmark with severe attribute imbalance.

KENTECH Ultra High Power Semiconductor Lab — Apr 2023 – Oct 2024
Undergraduate Researcher, Advisor: Prof. Hyemin Kang — *Naju, Korea*

- Characterized static (output, transfer, capacitance, breakdown) and dynamic (double-pulse test) behavior of 1200 V SiC MOSFETs and IGBTs.
- Measured $\sim$3$\times$ lower turn-on and $\sim$6$\times$ lower turn-off switching energy for the SiC MOSFET than for the IGBT.

Publications & Presentations

- **Yunui Kang**, Jiseung Kim, Yeseong Kim, Hyunju Kim, and Hyunwoo Lee, "HD-NIDSBench: A Benchmarking Framework for Efficient Intrusion Detection Systems Using Hyperdimensional Computing," *Conference on Information Security and Cryptography – Winter (CISC-W)*, Nov 2025. **(Oral)**
- Hakyeong Seong, **Yunui Kang**, Hyunju Kim, and Hyunwoo Lee, "PQNetSim: Simulator for Measuring PQC Networking Overhead," *Conference on Information Security and Cryptography – Summer (CISC-S)*, Jun 2025. (Poster)
- Yeonwoo Seo, Seoeun Choi, **Yunui Kang**, and Hyemin Kang, "Comparative Study on the Static and Dynamic Performance of MOSFET and IGBT Devices," *1st KENTECH–GIST Academic Exchange Symposium*, Dec 2024. (Poster)

Selected Projects

Trust Gateway: A Zero Trust Proxy for Dynamic Trust Evaluation and Re-authentication
Capstone Design Project with ETRI *Mar – Jun 2026*

- Co-designed a proxy that recomputes a per-request trust score (0–1) from 24 behavioral features aggregated into 6 interpretable signals.
- Mapped scores to Allow, Monitor, Re-authenticate, or Block and tuned signal weights via SLSQP; won the Outstanding Poster Award.

Frequency Stabilization Control Design for Turbine Systems
Control Theory and Applications *Spring 2025*

- Modeled a coupled steam–gas turbine system with 5% droop control in Simulink; before compensation, its phase margin was only 1.26°.
- Designed compensators achieving 40–60° phase margin and $\geq$6 dB gain margin, and added automatic generation control (AGC) to eliminate steady-state frequency error.

Constrained Logistic Regression via Interior-Point Method
Optimization Theory *Fall 2025*

- Built a log-barrier Newton solver that encodes medical domain knowledge as sign constraints on 30 feature weights.
- Verified the KKT conditions at the solution; test accuracy (98.25%) slightly exceeded training accuracy (97.80%), suggesting a regularizing effect.

Honors & Funding

President's Scholarship for Research (KRW 25M)	*KENTECH; funded visiting research at UT Dallas*	Sep 2026
Outstanding Poster Award	*IR/CAPS Research Showcase, KENTECH*	Jun 2026
Jeonnam Student Venture Startup Expedition	*Selected Team (KRW 10M funding)*	Mar 2026 – Present

Skills & Languages

Languages	:	Korean (native), English (TOEFL iBT 110, band score 5.5/6.0; TOEIC 980)
Programming	:	Python, C++
Security	:	OCPP 1.6, protocol fuzzing, TLS/DTLS 1.3, PQC (ML-KEM)
Tools	:	PyTorch, TensorFlow, Simulink, LaTeX, Linux tc network emulation, Raspberry Pi

Leadership & Service

Undergraduate Teaching Assistant	*English for Specific Purposes (ESP), KENTECH*	2024 – 2026
Class Representative	*Dept. of Energy Engineering (Leadership Scholarship, 2 semesters)*	Mar – Dec 2025
General Secretary	*Power Systems Academic Club, KENTECH*	Mar – Dec 2025

- Ran the club's regular study sessions and seminars on power grid control and operation.

Member	*Student Board of Audit and Inspection, KENTECH*	Apr – Dec 2024
Member	*Student Council, KENTECH*	Jul – Dec 2024
Working Scholar	*Technology Commercialization, Research Support, and IT Operations Teams*	2024 – 2026